Modeling Choices Document

Bayesian Smoothing Of Functional Data On Spatial Domains

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Dataset and Model

The dataset consists of observations of temperature on a 1-degree resolution grid over the Earth's surface. For each location s_k of the lattice ($k = 1, ..., N$), we have the average temperature for $T = 25$ consecutive years.

Let $Y_i(s)$ be the observed temperature at location $s \in \Omega$ (where Ω is the sphere $\mathbb{S}^2$) during year i. We model the underlying continuous temperature field as:

$$Y_i(s) = a_i + \sum_{j=1}^{V} \beta_{i,j}\varphi_j(s) + \varepsilon_i(s) \quad s \in \Omega$$

where β_i is the vector of coefficient for a given triangulation $\mathcal{T}_h$ of the domain Ω, V is the number of degrees of freedom of the given mesh, $\varphi_j(s)$ are the $\mathbb{P}^1$ FEM basis functions and $\varepsilon_i(s) \sim \mathcal{N}(0, \sigma_\varepsilon^2)$

We can then rewrite the vector form for all locations for year i as:

$$\boldsymbol{Y}_i = a_i \mathbf{1}_N + \Phi\boldsymbol{\beta}_i + \boldsymbol{\varepsilon}_i$$

where Φ is the $N \times V$ sparse observation matrix with entries $\Phi_{k,j} = \varphi_j(s_k)$

Model Interpretation

We assign a Gaussian prior to the spatial coefficient vectors $\boldsymbol{\beta}_i$, assuming that for each year these vectors are independent and identically distributed:

$$\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, ..., \boldsymbol{\beta}_{25} \overset{\text{iid}}{\sim} \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

By centering the coefficients around their mean, we can define the zero-mean anomaly $\tilde{\boldsymbol{\beta}}_i = \boldsymbol{\beta}_i - \boldsymbol{\mu}$ such that $\tilde{\boldsymbol{\beta}}_i \sim \mathcal{N}(\mathbf{0}, \Sigma)$. The model can then be explicitly rewritten to separate the mean spatial field from the yearly deviations:

$$\boldsymbol{Y}_i = a_i \mathbf{1}_N + \Phi\boldsymbol{\mu} + \Phi\tilde{\boldsymbol{\beta}}_i + \boldsymbol{\varepsilon}_i$$

where each term has a clear physical interpretation:

- a_i: the global temporal effect (intercept) for year i.
- $\Phi\boldsymbol{\mu}$: the long-term average temperature field evaluated at the observation locations.
- $\Phi\tilde{\boldsymbol{\beta}}_i$: the spatially correlated anomaly for year i, capturing how local temperatures deviate from the long-term mean.
- $\boldsymbol{\varepsilon}_i$: the unstructured measurement error at each location, assumed to be independent with $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2 I_N)$.

Bayesian Inference on the model

The SPDE Approach for the Spatial Covariance

When dealing with massive spatial datasets, computing the dense covariance matrix $\mathbf{\Sigma}$ and its inverse is computationally prohibitive. In our setting, we have $N \approx 64,800$ observations per year on the sphere. While projecting the continuous spatial field onto a discrete triangular mesh reduces the dimensionality to V vertices (where $V \ll N$), constructing and inverting a dense $V \times V$ covariance matrix still requires $\mathcal{O}(V^3)$ operations, creating a severe bottleneck for Markov Chain Monte Carlo (MCMC) algorithms.

To overcome this computational burden, we adopt the Stochastic Partial Differential Equation (SPDE) approach introduced by Lindgren et al. (2011). This framework establishes an explicit link between continuous spatial Gaussian fields with Matérn covariance functions and discrete Gaussian Markov Random Fields (GMRFs).

Instead of defining a dense covariance matrix directly, we define the spatial random field $u(s)$ as the stationary solution to the following linear SPDE:

$$(1 - \nabla \cdot \boldsymbol{H}(s)\nabla)(\tau u(s)) = \mathcal{W}(s)$$

where:

- $\mathcal{W}(s)$ is a Gaussian spatial white noise process on the spherical manifold $\mathcal{S}^2$.
- $\boldsymbol{H}(s)$ is a 2×2 positive-definite tensor field defined on the local tangent plane at each location s. While its spatial orientation rotates to follow the curvature of $\mathcal{S}^2$, its eigenvalues (representing the variance along the local East and North axes) are global parameters to be estimated.
- τ is a precision parameter controlling the marginal variance of the field.

Using the Finite Element Method (FEM), the continuous spatial field $u(s)$ is approximated as a linear combination of localized basis functions $\varphi_k(s)$, defined on the V vertices of the triangular mesh:

$$u(s) \approx \sum_{k=1}^{V} \varphi_k(s) u_k$$

Computing the weak (Galerkin) formulation of the SPDE requires integrating against test functions φ_j. Applying the divergence theorem on the manifold yields:

$$\tau \sum_{i=1}^{V} u_i \int_{\mathcal{S}^2} \big(\nabla \varphi_j(s)\big)^\top \boldsymbol{H}(s) (\nabla \varphi_i(s))\, \mathrm{d}s + \tau \sum_{i=1}^{V} u_i \int_{\mathcal{S}^2} \varphi_j(s)\varphi_i(s)\, \mathrm{d}s$$
$$= \int_{\mathcal{S}^2} \mathcal{W}(s)\varphi_j(s)\, \mathrm{d}s \quad \forall j \in \{1, ..., V\}$$

Focusing on the first integral, we decompose the local anisotropy tensor $\boldsymbol{H}(s)$ into the fixed geometric projection tensors along the local East (x) and North (y) axes:

$$\boldsymbol{H}(s) = h_1 \boldsymbol{H}_x(s) + h_2 \boldsymbol{H}_y(s)$$

with the scalar variances h_1 and h_2 being independent of the spatial location s. By substituting this into the stiffness integral and exploiting linearity, we can write:

$$\int_{\mathcal{S}^2} \big(\nabla \varphi_j(s)\big)^\top \boldsymbol{H}(s) (\nabla \varphi_i(s))\, \mathrm{d}s = h_1 \underbrace{\int_{\mathcal{S}^2} \big(\nabla \varphi_j(s)\big)^\top \boldsymbol{H}_{x(s)} (\nabla \varphi_i(s))\, \mathrm{d}s}_{(\boldsymbol{K}_x)_{ji}}$$
$$+ h_2 \underbrace{\int_{\mathcal{S}^2} \big(\nabla \varphi_j(s)\big)^\top \boldsymbol{H}_{y(s)} (\nabla \varphi_i(s))\, \mathrm{d}s}_{(\boldsymbol{K}_y)_{ji}}$$

where $(\boldsymbol{K}_x)_{ji}$ and $\big(\boldsymbol{K}_y\big)_{ji}$ are the (j, i)-th entries of the geometric stiffness matrices constructed purely along the local longitudinal and latitudinal directions, respectively.

Consequently, the global anisotropic stiffness matrix $\boldsymbol{K}_{\boldsymbol{H}}$ can be expressed as a simple linear combination of two fixed matrices:

$$\boldsymbol{K}_{\boldsymbol{H}} = h_1 \boldsymbol{K}_x + h_2 \boldsymbol{K}_y$$

Defining also the corresponding matrix and vectors:

$$\boldsymbol{u} = (u_1, u_2, ..., u_V)^\top,$$
$$\boldsymbol{C}_{ji} = \int_{\mathcal{S}^2} \varphi_{j(s)} \varphi_{i(s)}\, \mathrm{d}s,$$
$$\boldsymbol{f}_j = \int_{\mathcal{S}^2} \mathcal{W}(s) \varphi_{j(s)}\, \mathrm{d}s,$$

We can rewrite the SPDE in matrix form as:

$$\tau\big(h_1\boldsymbol{K}_x + h_2\boldsymbol{K}_y + \boldsymbol{C}\big)\boldsymbol{u} = \boldsymbol{f}$$

Computing the covariance between two elements of our vector $\boldsymbol{f}$:

$$\begin{aligned}
\operatorname{Cov}\big(f_i, f_j\big) &= E\big[f_i f_j\big] \\
&= E\left[\int_{\mathcal{S}^2} \mathcal{W}(s)\varphi_i(s)\,\mathrm{d}s \int_{\mathcal{S}^2} \mathcal{W}(s')\varphi_j(s')\,\mathrm{d}s'\right] \\
&= \int_{\mathcal{S}^2}\int_{\mathcal{S}^2} \underbrace{E[\mathcal{W}(s)\mathcal{W}(s')]}_{\delta(s-s')}\varphi_i(s)\varphi_j(s')\,\mathrm{d}s\,\mathrm{d}s' \\
&= \int_{\mathcal{S}^2} \varphi_i(s)\varphi_j(s)\,\mathrm{d}s = \boldsymbol{C}_{ij}
\end{aligned}$$

so applying the covariance to both sides:

$$\tau^2\big(h_1\boldsymbol{K}_x + h_2\boldsymbol{K}_y + \boldsymbol{C}\big)\boldsymbol{\Sigma}_u\big(h_1\boldsymbol{K}_x + h_2\boldsymbol{K}_y + \boldsymbol{C}\big)^\top = \boldsymbol{C}$$

Since the mass matrix $\boldsymbol{C}$ and the stiffness matrices $\boldsymbol{K}_x$ and $\boldsymbol{K}_y$ are symmetric by construction, the transpose operation leaves the matrices unchanged. We can then isolate the spatial covariance matrix $\boldsymbol{\Sigma}_u$:

$$\boldsymbol{\Sigma}_u = \frac{1}{\tau^2}\big(h_1\boldsymbol{K}_x + h_2\boldsymbol{K}_y + \boldsymbol{C}\big)^{-1}\boldsymbol{C}\big(h_1\boldsymbol{K}_x + h_2\boldsymbol{K}_y + \boldsymbol{C}\big)^{-1}$$

In our Bayesian framework, the Gibbs sampler requires the precision matrix $\boldsymbol{Q} = \boldsymbol{\Sigma}_u^{-1}$ rather than the covariance. Inverting $\boldsymbol{\Sigma}_u$ yields the analytical GMRF precision matrix:

$$\boldsymbol{Q} = \tau^2\big(h_1\boldsymbol{K}_x + h_2\boldsymbol{K}_y + \boldsymbol{C}\big)\boldsymbol{C}^{-1}\big(h_1\boldsymbol{K}_x + h_2\boldsymbol{K}_y + \boldsymbol{C}\big)$$

To optimize computational efficiency, the mass matrix $\boldsymbol{C}$ is commonly replaced by its lumped diagonal version $\tilde{\boldsymbol{C}}$, where $\tilde{C}_{ii} = \sum_j C_{ij}$. One can rigorously prove that this approximation is theoretically sound using the First Strang Lemma.

By treating the hyperparameters τ, h_1, and h_2 as random variables within our hierarchical Bayesian framework, we can assign appropriate prior distributions to them. This allows us to update the spatial dependency structure and sample the

latent field efficiently, entirely bypassing the need to construct or invert dense covariance matrices.

As demonstrated by Lindgren et al. (2011) constructing the GMRF precision matrix $\boldsymbol{Q}$ via finite element methods (using the exact mass matrix $\boldsymbol{C}$) provides an explicit, rigorous, and highly sparse representation of the inverse Matérn covariance matrix on the discrete mesh.

Priors and Full Conditional Distributions

To implement a highly efficient Gibbs sampler, we derive the full conditional distribution for each parameter block. To do that, we assume that our global parameters are **a priori independent**. This crucial assumption allows us to isolate the full conditional of any given parameter by simply multiplying the data likelihood by the specific prior of the parameter being updated, absorbing everything else into the proportionality constant.

Global temporal effect (a_i)

From Bayes' theorem, the full conditional distribution for the global temporal intercept of the i-th year is proportional to the joint posterior. We now use the conditional independence of the observations and the a priori independence of the parameters:

$$\mathcal{L}(a_i \mid \text{rest}) \propto \mathcal{L}(a_i) f(\boldsymbol{Y}_i \mid \text{rest})$$

Looking at our model, it is immediate to see that the observations for year i follow a multivariate normal distribution:

$$\boldsymbol{Y}_i \mid \text{rest} \sim \mathcal{N}\left(a_i \mathbf{1}_N + \Phi\boldsymbol{\mu} + \Phi\tilde{\boldsymbol{\beta}}_i, \sigma_\varepsilon^2 \boldsymbol{I}_N\right)$$

Let $\boldsymbol{r}_i = \Phi\boldsymbol{\mu} + \Phi\tilde{\boldsymbol{\beta}}_i$ be the vector of the spatial components. We can write the formula for our posterior kernel by expanding the quadratic form:

$$\begin{aligned}
\mathcal{L}(a_i \mid \text{rest}) &\propto \mathcal{L}(a_i) \exp\left[-\frac{1}{2\sigma_\varepsilon^2}(\boldsymbol{Y}_i - \boldsymbol{r}_i - a_i\mathbf{1}_N)^T(\boldsymbol{Y}_i - \boldsymbol{r}_i - a_i\mathbf{1}_N)\right] \\
&\propto \mathcal{L}(a_i) \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left(N a_i^2 - 2a_i \sum_{k=1}^{N}(Y_{i,k} - r_{i,k})\right)\right] \\
&\propto \mathcal{L}(a_i) \exp\left[-\frac{1}{2} q_{\text{data}}(a_i - m_{\text{data}})^2\right]
\end{aligned}$$

where

$$q_{\text{data}} = \frac{N}{\sigma_\varepsilon^2} \quad m_{i,\text{data}} = \frac{1}{N}\sum_{k=1}^{N}\left(Y_{i,k} - r_{i,k}\right)$$

By choosing a normal prior $a_i \sim \mathcal{N}\left(m_{\text{prior}}, q_{\text{prior}}^{-1}\right)$, we recognize a standard univariate **Normal-Normal** conjugate update. The full conditional posterior is $a_i \mid \text{rest} \sim \mathcal{N}\left(a_{i,\text{post}}, q_{\text{post}}^{-1}\right)$, where:

$$q_{\text{post}} = q_{\text{prior}} + q_{\text{data}} \quad a_{i,\text{post}} = q_{\text{post}}^{-1}\left(q_{\text{prior}} m_{\text{prior}} + q_{\text{data}} m_{i,\text{data}}\right)$$

Measurement error variance (σ_ε^2)
Similarly, isolating the variance parameter:

$$\boldsymbol{\varepsilon}_i = \mathbf{Y}_i - a_i \mathbf{1}_N - \Phi\left(\boldsymbol{\mu} + \tilde{\boldsymbol{\beta}}_i\right) \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

Crucially, while the specific realizations of the measurement error $\boldsymbol{\varepsilon}_i$ differ from year to year, we assume the observation process is homoscedastic over time. This means the variance parameter σ_ε^2 is shared across all $M = 25$ years. The combined likelihood for σ_ε^2 thus incorporates all spatial observations across all years. We assume our parameters to be a priori independent so that:

$$\mathcal{L}(\sigma_\varepsilon^2 \mid \text{rest}) \propto f(\text{rest} \mid \sigma_\varepsilon^2)\mathcal{L}(\sigma_\varepsilon^2) \propto f(\varepsilon_i \mid \sigma_\varepsilon^2)\mathcal{L}(\sigma_\varepsilon^2)$$

Since we are pooling all our available data we have that this likelihood is comprised of N independent and identically distributed samples from a normal distribution, so:

$$\begin{aligned} f(\varepsilon_i \mid \sigma_\varepsilon^2) &= \prod_{i=1}^{M}\prod_{k=1}^{N} \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\varepsilon_i^2(s_k)\right] \\ &= \left(2\pi\sigma_\varepsilon^2\right)^{-MN/2} \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\sum_{i=1}^{M}\boldsymbol{\varepsilon}_i^T\boldsymbol{\varepsilon}_i\right] \end{aligned}$$

in the computation of our posterior we have to take all of our $\varepsilon_i(s_k)$ as given constants:

$$\mathcal{L}(\sigma_\varepsilon^2 \mid \text{rest}) \propto \left(\sigma_\varepsilon^2\right)^{-MN/2} \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\sum_{i=1}^{M}\boldsymbol{\varepsilon}_i^T\boldsymbol{\varepsilon}_i\right]\mathcal{L}(\sigma_\varepsilon^2)$$

Looking at this formula it's clear that if we take an inverse-gamma distribution as a prior we have conjugacy, so $\sigma_\varepsilon^2 \sim \text{Inv-Gamma}(a, b)$, with a and b chosen to

have a non informative prior since the large number of data available will be more than enough to do inference, this choice leads to the full conditional being:

$$\sigma_\varepsilon^2 \mid \text{rest} \sim \text{Inv-Gamma}\big(a_{\text{post}}, b_{\text{post}}\big)$$

with values for a_{post} and b_{post} being:

$$a_{\text{post}} = \frac{MN}{2} + a, \quad b_{\text{post}} = \frac{1}{2}\sum_{i=1}^{M} \boldsymbol{\varepsilon}_i^T \boldsymbol{\varepsilon}_i + b$$

Global temperature mean field ($\boldsymbol{\mu}$)

Following Bayes' theorem and the a priori independence assumption, the full conditional for the spatial mean vector $\boldsymbol{\mu}$ is proportional to the joint data likelihood multiplied by the prior:

$$\mathcal{L}(\boldsymbol{\mu} \mid \text{rest}) \propto f(\boldsymbol{Y} \mid \text{rest})\mathcal{L}(\boldsymbol{\mu})$$

The distribution of $\boldsymbol{Y} \mid \text{rest}$ can be computed directly from the original formula for our model, considering that, as with the variance of the error, we observe $M = 25$ independent identically distributed realizations of $\boldsymbol{Y}$

$$\boldsymbol{Y}_i = a_i \mathbf{1}_N + \Phi\boldsymbol{\mu} + \Phi\tilde{\boldsymbol{\beta}}_i + \boldsymbol{\varepsilon}_i \sim \mathcal{N}\big(\boldsymbol{r_i} + \Phi\boldsymbol{\mu}, \sigma_\varepsilon^2 \boldsymbol{I}_N\big)$$

where $\boldsymbol{r}_i = a_i \mathbf{1}_N + \Phi\tilde{\boldsymbol{\beta}}_i$. Using this formula to compute the posterior we have:

$$\begin{aligned}
\mathcal{L}(\boldsymbol{\mu} \mid \text{rest}) &\propto \mathcal{L}(\boldsymbol{\mu}) \prod_{i=1}^{M} \exp\left[-\frac{1}{2\sigma_\varepsilon^2}(\boldsymbol{Y_i} - \boldsymbol{r_i} - \Phi\boldsymbol{\mu})^T(\boldsymbol{Y_i} - \boldsymbol{r_i} - \Phi\boldsymbol{\mu})\right] \\
&\propto \mathcal{L}(\boldsymbol{\mu}) \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left(M\boldsymbol{\mu}^T\Phi^T\Phi\boldsymbol{\mu} - 2\sum_{i=1}^{M}(\boldsymbol{Y}_i - \boldsymbol{r}_i)^T\Phi\boldsymbol{\mu}\right)\right] \\
&\propto \mathcal{L}(\boldsymbol{\mu}) \exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left(\boldsymbol{\mu}^T(M\Phi^T\Phi)\boldsymbol{\mu} - 2\sum_{i=1}^{M}(\boldsymbol{Y}_i - \boldsymbol{r}_i)^T\Phi\boldsymbol{\mu}\right)\right] \\
&\propto \mathcal{L}(\boldsymbol{\mu}) \exp\left[-\frac{1}{2}(\boldsymbol{\mu} - \boldsymbol{m}_{\text{data}})^T\boldsymbol{Q}_{\text{data}}(\boldsymbol{\mu} - \boldsymbol{m}_{\text{data}})\right]
\end{aligned}$$

where:

$$\boldsymbol{m}_{\text{data}} = \frac{1}{M}\left(\Phi^T\Phi\right)^{-1}\sum_{i=1}^{M}\Phi^T(\boldsymbol{Y}_i - \boldsymbol{r}_i) \quad \boldsymbol{Q}_{\text{data}} = \frac{M}{\sigma_\varepsilon^2}\Phi^T\Phi$$

This is the kernel of a normal distribution, so by choosing: $\boldsymbol{\mu} \sim \mathcal{N}\left(\boldsymbol{\mu}_{\text{prior}}, \boldsymbol{Q}_{\text{prior}}^{-1}\right)$ we have a **Normal-Normal** model. The full conditional posterior is $\boldsymbol{\mu} \mid \text{rest} \sim \mathcal{N}\left(\boldsymbol{\mu}_{\text{post}}, \boldsymbol{Q}_{\text{post}}^{-1}\right)$, where the posterior precision matrix is elegantly given by the sum of the prior and data precisions:

$$\boldsymbol{Q}_{\text{post}} = \boldsymbol{Q}_{\text{prior}} + \boldsymbol{Q}_{\text{data}}$$

and the posterior mean vector is precision-weighted:

$$\boldsymbol{\mu}_{\text{post}} = \boldsymbol{Q}_{\text{post}}^{-1}\left(\boldsymbol{Q}_{\text{prior}}\boldsymbol{\mu}_{\text{prior}} + \boldsymbol{Q}_{\text{data}}\boldsymbol{m}_{\text{data}}\right)$$

Spatiotemporal latent fields ($\tilde{\boldsymbol{\beta}}_i$)

As we already did, we can compute the posterior using the likelihood of our data $\boldsymbol{Y}_i$ given the rest. Since all of our parameters are a priori independent and the observations are conditionally independent across years, the full conditional effectively isolates the single-year likelihood:

$$\mathcal{L}\left(\tilde{\boldsymbol{\beta}}_i \mid \text{rest}\right) \propto \mathcal{L}\left(\tilde{\boldsymbol{\beta}}_i\right) f(\boldsymbol{Y}_i \mid \text{rest})$$

We already saw that $\boldsymbol{Y}_i \mid \text{rest} \sim \mathcal{N}\left(a_i\mathbf{1}_N + \Phi\boldsymbol{\mu} + \Phi\tilde{\boldsymbol{\beta}}_i, \sigma_\varepsilon^2\boldsymbol{I}_N\right)$. Calling $\boldsymbol{r}_i = a_i\mathbf{1}_N + \Phi\boldsymbol{\mu}$, we can write the formula for our posterior kernel:

$$\begin{aligned}
\mathcal{L}\left(\tilde{\boldsymbol{\beta}}_i \mid \text{rest}\right) &\propto \mathcal{L}\left(\tilde{\boldsymbol{\beta}}_i\right)\exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left(\boldsymbol{Y}_i - \boldsymbol{r}_i - \Phi\tilde{\boldsymbol{\beta}}_i\right)^T\left(\boldsymbol{Y}_i - \boldsymbol{r}_i - \Phi\tilde{\boldsymbol{\beta}}_i\right)\right] \\
&\propto \mathcal{L}\left(\tilde{\boldsymbol{\beta}}_i\right)\exp\left[-\frac{1}{2\sigma_\varepsilon^2}\left(\tilde{\boldsymbol{\beta}}_i^T\Phi^T\Phi\tilde{\boldsymbol{\beta}}_i - 2(\boldsymbol{Y}_i - \boldsymbol{r}_i)^T\Phi\tilde{\boldsymbol{\beta}}_i\right)\right] \\
&\propto \mathcal{L}\left(\tilde{\boldsymbol{\beta}}_i\right)\exp\left[-\frac{1}{2}\left(\tilde{\boldsymbol{\beta}}_i - \tilde{\boldsymbol{\beta}}_{i,\text{data}}\right)^T\boldsymbol{Q}_{\text{data}}\left(\tilde{\boldsymbol{\beta}}_i - \tilde{\boldsymbol{\beta}}_{i,\text{data}}\right)\right]
\end{aligned}$$

where the data precision matrix and the data mean vector for the i-th year are:

$$\boldsymbol{Q}_{\text{data}} = \frac{1}{\sigma_\varepsilon^2}\Phi^T\Phi \quad \tilde{\boldsymbol{\beta}}_{i,\text{data}} = \left(\Phi^T\Phi\right)^{-1}\Phi^T(\boldsymbol{Y}_i - \boldsymbol{r}_i)$$

From the SPDE approach we mentiond earlier we derive a normal distribution with parameters $\mathcal{L}\left(\tilde{\boldsymbol{\beta}}_i\right) = \mathcal{N}\left(\mathbf{0}, \boldsymbol{Q}_{\text{prior}}^{-1}(\tau, h_1, h_2)\right)$ obtaining a **Normal-Normal** conjugate update.

The full conditional posterior is $\tilde{\boldsymbol{\beta}}_i \mid \text{rest} \sim \mathcal{N}\left(\tilde{\boldsymbol{\beta}}_{i,\text{post}}, \boldsymbol{Q}_{\text{post}}^{-1}\right)$, with precision and mean updated as:

$$\boldsymbol{Q}_{\text{post}} = \boldsymbol{Q}_{\text{prior}} + \boldsymbol{Q}_{\text{data}}$$
$$\tilde{\boldsymbol{\beta}}_{i,\text{post}} = \boldsymbol{Q}_{\text{post}}^{-1}\left(\boldsymbol{Q}_{\text{data}}\tilde{\boldsymbol{\beta}}_{i,\text{data}}\right)$$

SPDE Hyperparameters ($\boldsymbol{\theta}$)

The spatial precision matrix $\boldsymbol{Q}$ is completely determined by the hyperparameters $\boldsymbol{\theta} = (\tau^2, h_1, h_2)$ via the specific anisotropic SPDE formulation:

$$\boldsymbol{Q}(\boldsymbol{\theta}) = \tau^2\left(h_1\boldsymbol{K}_x + h_2\boldsymbol{K}_y + \boldsymbol{C}\right)\boldsymbol{C}^{-1}\left(h_1\boldsymbol{K}_x + h_2\boldsymbol{K}_y + \boldsymbol{C}\right)$$

We assume $\boldsymbol{\theta}$ is independent with respect to all other parameters but $\tilde{\boldsymbol{\beta}}_i$, and as such:

$$\begin{aligned} f(\text{rest} \mid \boldsymbol{\theta}) &= f\left(\tilde{\boldsymbol{\beta}}_i \mid \boldsymbol{\theta}, a_i, \boldsymbol{\mu}, \sigma_\varepsilon^2\right)\mathcal{L}(a_i, \boldsymbol{\mu}, \sigma_\varepsilon^2 \mid \boldsymbol{\theta}) \\ &= f\left(\tilde{\boldsymbol{\beta}}_i \mid \boldsymbol{\theta}, a_i, \boldsymbol{\mu}, \sigma_\varepsilon^2\right)\mathcal{L}(a_i, \boldsymbol{\mu}, \sigma_\varepsilon^2) \\ &= f\left(\tilde{\boldsymbol{\beta}}_i \mid \boldsymbol{\theta}\right)\mathcal{L}(a_i, \boldsymbol{\mu}, \sigma_\varepsilon^2) \end{aligned}$$

since we are conditioning with respect to $\boldsymbol{\theta}$ we can simply throw away all terms that do not depend on $\boldsymbol{\theta}$:

$$\mathcal{L}(\boldsymbol{\theta} \mid \text{rest}) \propto \mathcal{L}(\boldsymbol{\theta}) \prod_{i=1}^{M} f\left(\tilde{\boldsymbol{\beta}}_i \mid \boldsymbol{\theta}\right)$$

where $f\left(\tilde{\boldsymbol{\beta}}_i \mid \boldsymbol{\theta}\right)$ is the multivariate normal density $\mathcal{N}\left(\mathbf{0}, \boldsymbol{Q}(\boldsymbol{\theta})^{-1}\right)$. Expanding this density explicitly:

$$\mathcal{L}(\boldsymbol{\theta} \mid \text{rest}) \propto |\boldsymbol{Q}(\boldsymbol{\theta})|^{\frac{M}{2}} \exp\left[-\frac{1}{2}\sum_{i=1}^{M} \tilde{\boldsymbol{\beta}}_i^T \boldsymbol{Q}(\boldsymbol{\theta})\tilde{\boldsymbol{\beta}}_i\right]\pi(\boldsymbol{\theta})$$

The hyperparameters enter the precision matrix non-linearly so finding a conjugate prior is unfeasible, so to sample from the posterior distribution we'll use **Metropolis-Within-Gibbs**. We need to choose the prior distribution for our parameter $\boldsymbol{\theta}$. Since all

the terms inside this vector need to be positive, we choose a weakly informative Log-Normal distribution for each component. Assuming a priori independence, we define:

$$\tau^2 \sim \text{LogNormal}(0, \sigma_\tau^2)$$
$$h_1 \sim \text{LogNormal}\left(0, \sigma_{h_1}^2\right)$$
$$h_2 \sim \text{LogNormal}\left(0, \sigma_{h_2}^2\right)$$

By setting the mean parameters to zero and choosing a sufficiently large variance, we ensure the prior remains flat. This weakly informative specification guarantees that the massive amount of observational data entirely drives the posterior inference.

Beyond statistical modeling, this choice offers a crucial computational advantage for our block Metropolis-Hastings update. To allow the algorithm to freely explore the parameter space without proposing invalid negative values, we reparameterize the problem in the logarithmic space:

$$\boldsymbol{\eta} = (\log \tau^2, \log h_1, \log h_2)$$

Under the independent Log-Normal assumption for $\boldsymbol{\theta}$, the prior for this transformed vector becomes a simple Multivariate Normal, $\boldsymbol{\eta} \sim \mathcal{N}\left(\boldsymbol{\mu}_\eta, \Sigma_\eta\right)$, with a diagonal covariance matrix. This transformation allows us to perform a Random Walk Metropolis step on the unconstrained space $\mathbb{R}^3$.